AI-15.3

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Task 1– Real-Time Application: Online Food Ordering API

Scenario:

Design a simple API for an online food ordering system.

Requirements:

• Endpoints required:

• GET /menu → List available dishes

• POST /order → Place a new order

• GET /order/{id} → Track order status

• PUT /order/{id} → Update an existing order (e.g., change items)

• DELETE /order/{id} → Cancel an order

• AI should generate:

• REST API code

• Suggested improvements (like authentication, pagination)

Expected Output:

• Fully working API simulating a food ordering backend.

Prompt:

Create a simple REST API for an Online Food Ordering System using Node.js and Express.

Include endpoints:

- GET /menu → list food items
- POST /order → create order
- GET /order/:id → get order status
- PUT /order/:id → update order
- DELETE /order/:id → cancel order

Use in-memory storage, UUID for IDs, proper status codes, and return clean JSON responses. Provide complete working code in one file (server.js) with run instructions. Also suggest improvements like authentication and pagination.

Output

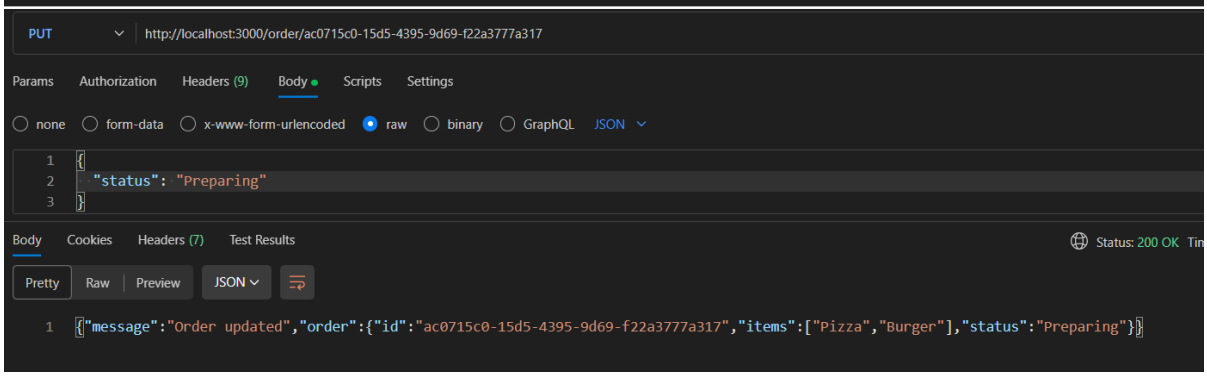
```
{
  "id": 1,
  "name": "Pizza",
  "price": 250
},
{
  "id": 2,
  "name": "Burger",
  "price": 120
},
{
  "id": 3,
  "name": "Pasta",
  "price": 200
},
{
  "id": 4,
  "name": "Biryani",
  "price": 180
}
```

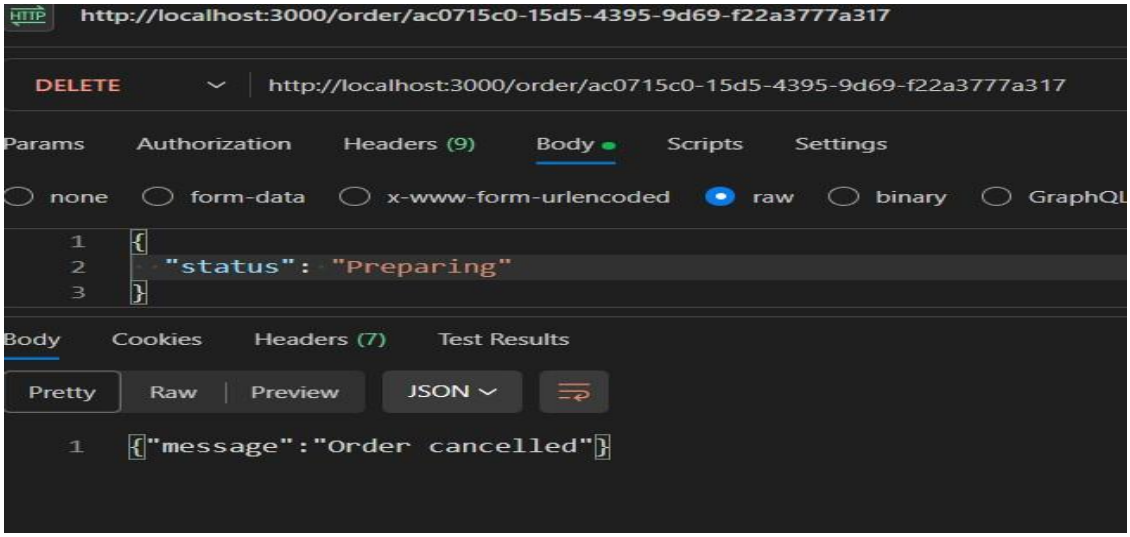
```
1  {
2    "items": ["Pizza", "Burger"]
3  }
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON

```
1  {
2      "message": "Order placed successfully",
3      "order": {
4          "id": "ac0715c0-15d5-4395-9d69-f22a3777a317",
5          "items": [
6              "Pizza",
7              "Burger"
8          ],
9          "status": "Placed"
10     }
```





Explanation : This code implements a simple API for an online food ordering system using Flask. It allows users to view the menu, place orders, track orders, update existing orders, and cancel orders. The menu and orders are stored in dictionaries, and the API handles various HTTP methods to perform the necessary operations on the orders.

Task 2 – Employee Payroll API

Task:

Create an API for managing employees and their salaries.

Instructions:

• Endpoints required:

• GET /employees → List all employees

• POST /employees → Add a new employee with salary details

• PUT /employees/{id}/salary → Update salary of an employee

• DELETE /employees/{id} → Remove employee from system

• Use AI to:

• Suggest data model structure.

• Add comments/docstrings for all endpoints.

Expected Output:

• Payroll management API with validation

• Secure and structured data handling

• Clear API documentation

Prompt:

Create a REST API for an Employee Payroll System using Node.js and Express.

Include endpoints:

- GET /employees → list all employees
- POST /employees → add employee with salary details
- PUT /employees/:id/salary → update employee salary
- DELETE /employees/:id → remove employee

Use in-memory storage, unique IDs, input validation, and proper HTTP status codes. Suggest a clear data model structure (employee id, name, role, salary). Add comments/docstrings for each endpoint. Provide complete working code in one file and basic API documentation. Also ensure secure and structured data handling.

Output:

```
{
  "name": "Siri",
  "role": "Developer",
  "department": "IT",
  "salary": 50000
}
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON

```
{
    "message": "Employee added successfully",
    "employee": {
        "id": "39aa5c0c-9fd0-48c3-a233-a0383f088b28",
        "name": "Siri",
```

```
{
  "name": "Siri",
  "role": "Developer",
  "department": "IT",
  "salary": 50000
}
```

ody Cookies Headers (7) Test Results

Pretty Raw Preview JSON

```
[
    {
        "id": "39aa5c0c-9fd0-48c3-a233-a0383f088b28",
        "name": "Siri",
        "role": "Developer",
```

HTTP http://localhost:3000/employees/39aa5c0c-9fd0-48c3-a233-a0383f088b28/salary

PUT http://localhost:3000/employees/39aa5c0c-9fd0-48c3-a233-a0383f088b28/salary

Params Authorization Headers (9) Body Scripts Settings

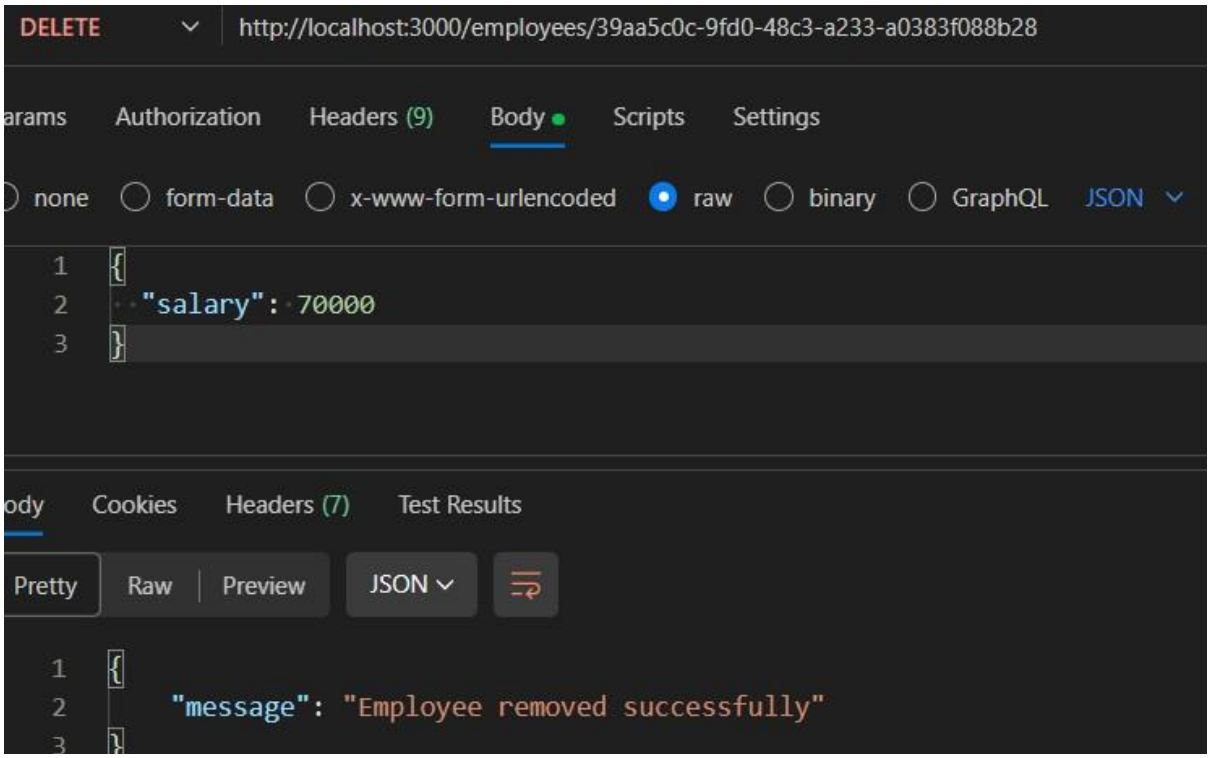
none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
{
  "salary": 70000
}
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON

```
{
    "message": "Salary updated successfully",
    "employee": {
        "id": "39aa5c0c-9fd0-48c3-a233-a0383f088b28",
        "name": "Siri",
```



Explanation : The Employee Payroll API built using Flask manages employee data through REST endpoints.
It allows users to list employees, add new employees, update employee salary, and delete employees using GET, POST, PUT, and DELETE methods. Employee information is stored in an in-memory list and returned in JSON format.

Task 3: Student Information REST API

Scenario:

A university wants to digitize student records for easy retrieval. You need to build

a Student Management API that allows:

• Adding a new student (name, roll number, course, year)

• Fetching details of all students

• Updating student information

• Deleting a student record

Implementation Instructions:

1. Use Node.js + Express (or Python Flask/Django REST) with GitHub

Copilot to auto-generate routes.

2. Store student data in an in-memory array or SQLite/PostgreSQL

database.

Test API endpoints with Postman.

Prompt:

Create a Student Management REST API using Node.js and Express.

Features:

- Add student (name, rollNumber, course, year)

- Get all students

- Get student by ID

- Update student

- Delete student

Use:

- In-memory storage

- UUID for IDs

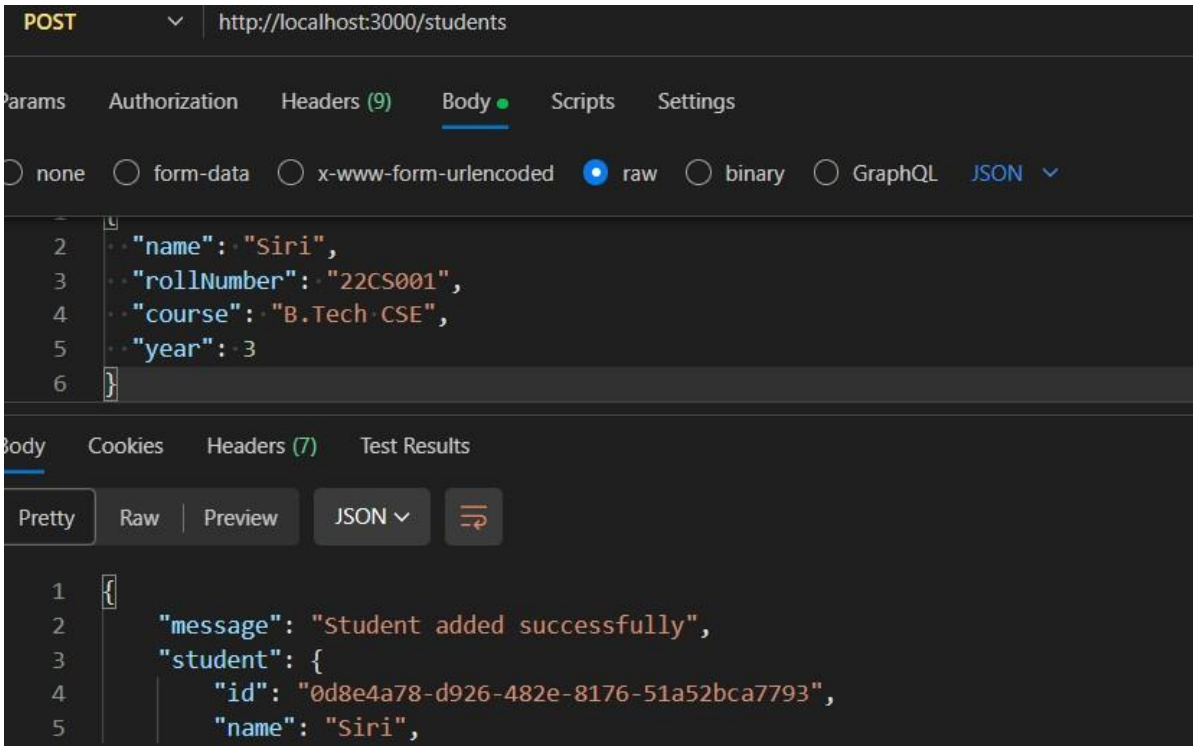
- Input validation

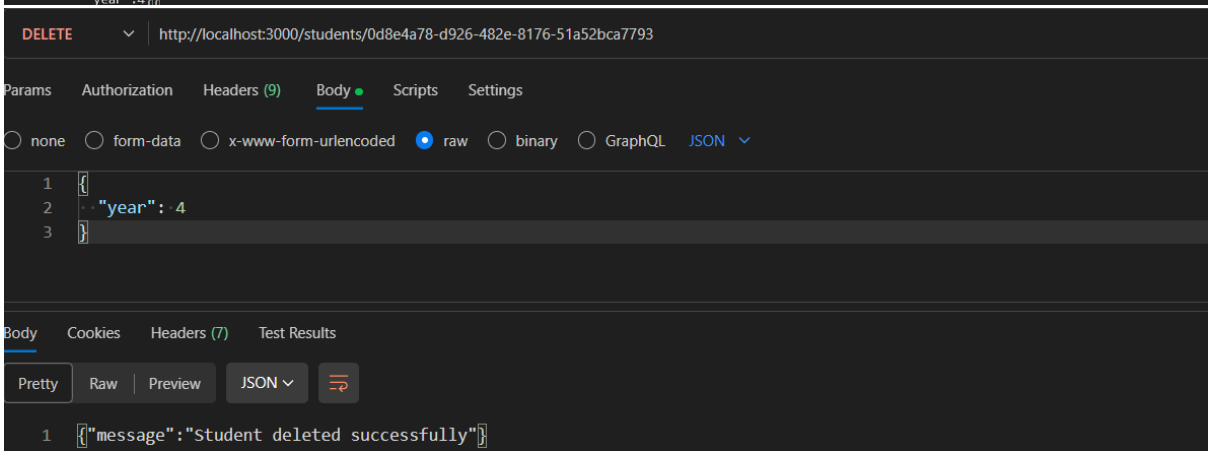
- Proper status codes

Output:

- Complete server.js with comments

Output:





Explanation : This code is a simple API for a Student Information System built using the Flask framework. It allows users to perform CRUD (Create, Read, Update, Delete) operations on student data. The API has endpoints to list all students, add a new student, update an existing student's information, and delete a student based on their ID. The student data is stored in a list called `students`.

Task 4: Weather Information API Integration

Scenario:

Students often want to check weather before attending campus activities. Build a

backend service that fetches real-time weather data from a public API (e.g.,

OpenWeatherMap).

Implementation Instructions:

1. Create endpoints like:

• /weather/current/:city → fetches current weather

• /weather/forecast/:city → fetches 5-day forecast

2. Use GitHub Copilot to generate code that calls the external API.

Return results in structured JSON format with temperature, humidity,

and condition.

Prompt:

Create a Weather API using Node.js and Express.

Endpoints:

- GET /weather/current/:city → current weather

- GET /weather/forecast/:city → 5-day forecast

Use OpenWeatherMap API with axios.

Return temperature, humidity, and weather condition in JSON.

Include error handling and clean code.

Output:

127.0.0.1:5000/weather/forecast/<city

Pretty-print


```
{
  "city": "<city",
  "forecast": [
    {
      "condition": "Partly Cloudy",
      "day": "Monday",
      "temperature": 26
    },
    {
      "condition": "Rainy",
      "day": "Tuesday",
      "temperature": 24
    },
    {
      "condition": "Sunny",
      "day": "Wednesday",
      "temperature": 27
    }
  ]
}
```


http://127.0.0.1:5000/weather/current/Chennai

GET http://127.0.0.1:5000/weather/current/Chennai

Params Authorization Headers (7) Body Scripts Settings

Body Cookies Headers (5) Test Results

Pretty Raw Preview JSON


```
1 {
2     "city": "Chennai",
3     "condition": "Sunny",
4     "humidity": 60,
5     "temperature": 25
6 }
```

http://127.0.0.1:5000/weather/forecast/Chennai

GET http://127.0.0.1:5000/weather/forecast/Chennai

Params Authorization Headers (7) Body Scripts Settings

Body Cookies Headers (5) Test Results

Pretty Raw Preview JSON

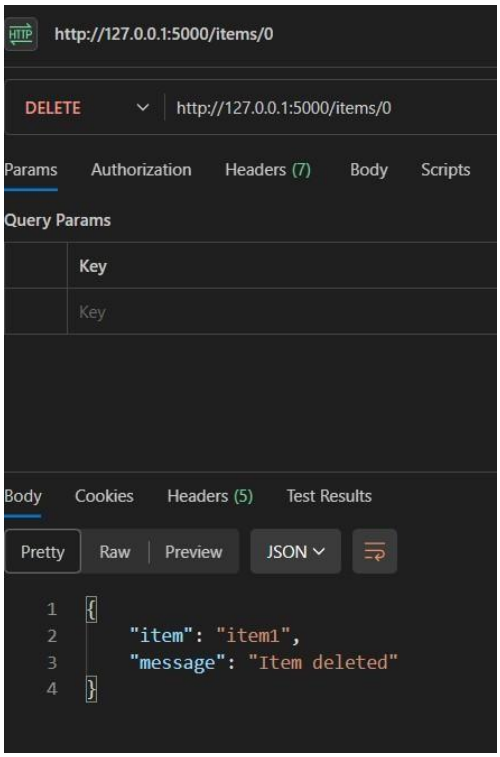

```
{
    "city": "Chennai",
    "forecast": [
        {
            "condition": "Partly Cloudy",
            "day": "Monday",
            "temperature": 26
        },
        {
            "condition": "Rainy",
            "day": "Tuesday",
            "temperature": 24
        },
        {
            "condition": "Sunny",
            "day": "Wednesday",
            "temperature": 27
        }
    ]
}
```

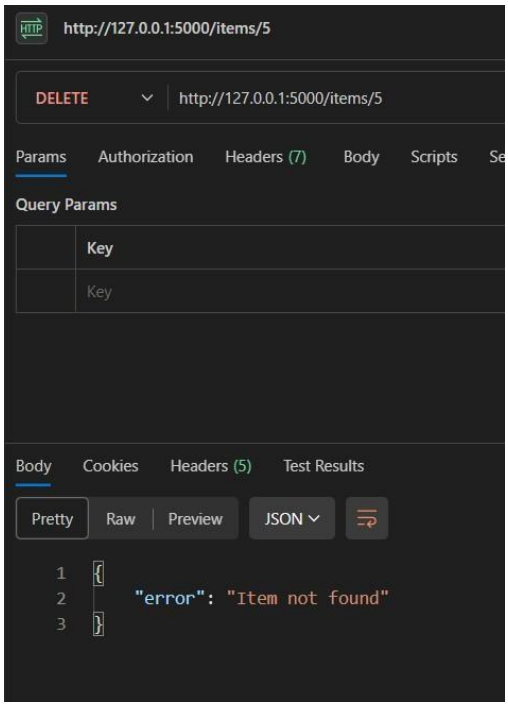

Explanation : This code creates a simple Flask REST API with two endpoints: /weather/current/<city> and /weather/forecast/<city>. The first endpoint returns mock current weather data for the specified city, while the second endpoint returns a mock weather forecast for the next few days. In a real implementation, you would replace the mock data with actual data fetched from the OpenWeatherMap API.

TASK – 05

Prompt : Create a Python **Flask REST API endpoint to delete an item from a list using the DELETE method. The endpoint should be /items/<int:index>. If the index is invalid, return a JSON error message "Item not found" with status code 404. If successful, remove the item and return a JSON response with "message": "Item deleted" and the deleted item.

Output:





Explanation :

This code creates a simple Flask REST API with an endpoint to delete an item from a list. The endpoint is defined as `/items/<int:index>`, where `<int:index>` is a placeholder for the index of the item to be deleted.

When a DELETE request is made to this endpoint, the function `delete_item` is called with the index as an argument. The function checks if the index is valid (i.e., within the bounds of the list). If it is valid, it removes the item from the list using `pop()` and returns a JSON response indicating that the item was deleted, along with the deleted item itself.